STUDENT MANAGER SYSTEM

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PROJECT TITLE:

JAVA SWING STUDENT MANAGER SYSTEM

PROBLEM STATEMENT: Managing student records manually is inefficient, error-prone, and time-consuming. This system automates student data handling using a GUI-based approach.

TARGET USERS:

School/College administrators
Teachers
Lab instructors

SYSTEM DESIGN

High-Level Architecture:

- MVC-like structure:
  - Model → Student class (ID, Name, Course)
  - View → Java Swing GUI (JFrame, JTable)
  - Controller → Action listeners (Add, Update, Delete)

Key Classes & Responsibilities:

- `Student` → Stores student data
- `StudentManagerGUI` → Handles UI and logic
- `DefaultTableModel` → Manages table data dynamically

Package Structure:

- `model` → Student class
- `view` → GUI components
- `controller` → Event handling logic

OOP CONCEPTS USED

Abstraction:

- Hidden implementation inside GUI event handlers
  - User interacts only with buttons and fields

Inheritance:

- StudentManagerGUI extends JFrame

Polymorphism:

- Event handling using overridden `actionPerformed()` method
  - Same interface used for multiple button actions

Design Pattern:

- MVC-inspired architecture

(Model-View-Controller separation)

TECHNICAL HIGHLIGHT

Key Design Decision:

Only product specifically dedicated to this niche market

Why this approach?

- Easy row addition and deletion
- No need for manual table refresh
- Real-time UI update support

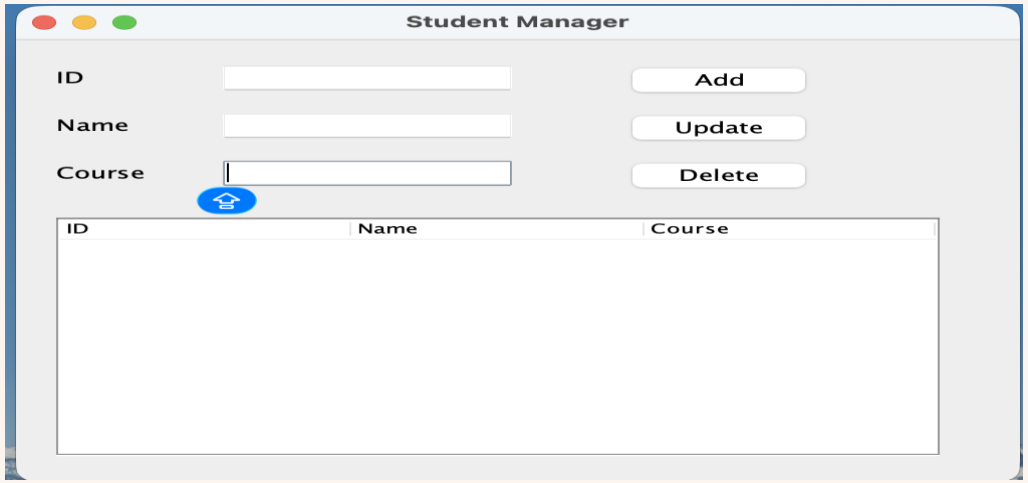
Additional Feature:

- Input validation (prevents empty ID or name entry)
- Instant table update after each operation

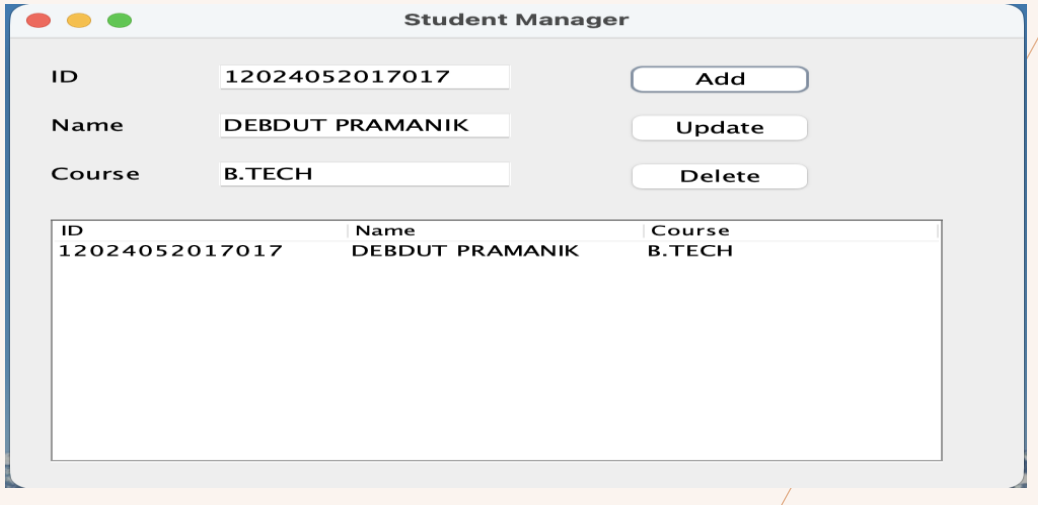
LIVE DEMO

Demo Scenario:

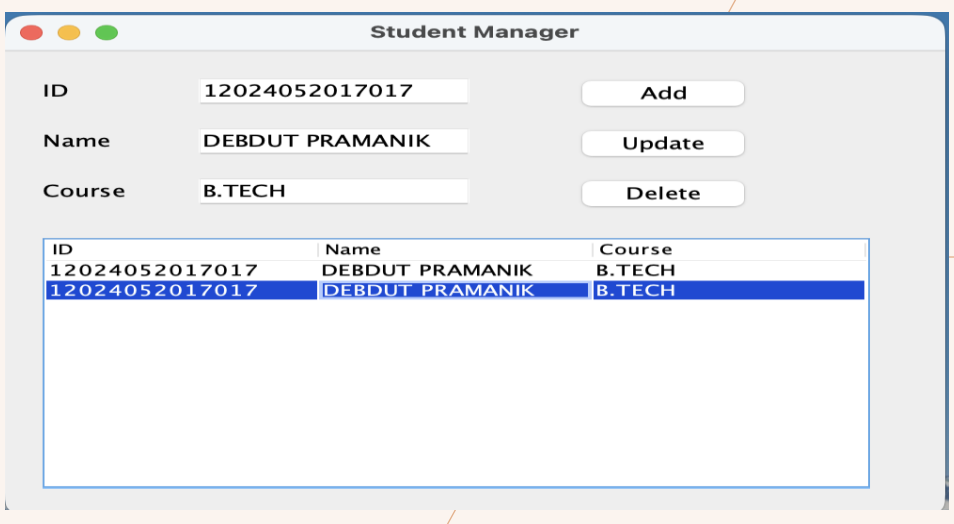
- Open application window



- Enter Student ID, Name, Course
- Click “Add” → Data appears in table



- SELECT ROW → CLICK “UPDATE” → MODIFY DETAILS
- CLICK “DELETE” → REMOVE RECORD



•Error Handling Demo:

•Try adding empty fields → System shows error message

•Invalid selection → Safe handling without crash

•Key Point to Say:

•“This system demonstrates real-time CRUD operations using Java Swing GUI with proper event handling.”

THANK YOU

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